Estimating the Sampling-Error Covariance Matrix over Time for Multivariate Fay–Herriot Poverty Mapping

Cross-time covariances of the direct estimators and how the EU SAE package estimates them

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1. Purpose and motivation

Multivariate Fay–Herriot (MFH) models extend the original area-level Fay–Herriot model (Fay and Herriot, 1979) by modeling several related direct estimators jointly. For poverty mapping over time, those related estimators are usually direct estimates for the same area in different survey years. For each area or domain, the model needs a sampling-error variance-covariance matrix for those direct estimates across time. The diagonal elements are the sampling variances of the direct estimates in each year. The off-diagonal elements are the cross-time covariances of the direct estimators.

For two years, the area-level sampling-error matrix supplied to the MFH model has the form

$$D_j = \begin{pmatrix} Var\left(e_{j1}\right) & Cov\left(e_{j1}, e_{j2}\right) \\ Cov\left(e_{j2}, e_{j1}\right) & Var\left(e_{j2}\right) \end{pmatrix},$$

where j indexes the area and e_{jt} is the sampling error of the direct estimator in year t. Benavent and Morales (2016) present the multivariate Fay–Herriot framework, in which the sampling-error covariance matrix is treated as a known model input, while Rao and Molina (2015) present the broader small area estimation framework for using direct estimators, sampling variances, EBLUPs, and MSEs in area-level SAE.

The off-diagonal covariance matters because the same survey design features that create dependence across years also affect the uncertainty of estimated changes over time. Cross-time covariance is especially relevant when surveys reuse primary sampling units (PSUs), use rotating panels, contain panel household components, or otherwise overlap clusters across waves. Ignoring such covariance can distort uncertainty measures for both direct changes and MFH-based change estimates.

In the EU SAE package, this matrix is not estimated freely for every area. Because area-level covariances are unstable when only a few sampled clusters overlap across waves, the package by default keeps the diagonal sampling variances area-specific but replaces the off-diagonal with a single national cross-time autocorrelation applied to those variances. Sections 3 to 6 establish what the off-diagonal represents and why the survey design identifies it; Section 8 sets out the simplification that is actually used.

QER implementation scope. This note concerns covariance of Direct sampling errors, not MCPE of model prediction errors. The latter and annual MSE are estimated jointly by the final-estimator bootstrap described in the companion MSE note. The implementation checks and simulation evidence are documented in docs/guidance/METHODOLOGY_ALIGNMENT.md and

docs/validation/MCPE_VALIDATION_STATUS.md. Numerical agreement and Monte Carlo precision do not certify interval coverage for all survey designs. Review model assumptions, boundary sensitivity and any undercoverage before dissemination. Use the latest validation status for current experiment counts and results.

2. Two covariance matrices in MFH

MFH models contain two conceptually different covariance matrices.

The first is the random-effects covariance matrix. This matrix is estimated by the MFH model and describes the relationship among area-level random effects across outcomes or time periods. Different structures for this matrix define model variants such as MFH1, MFH2, and MFH3. Benavent and Morales (2016) provide the methodological basis for these multivariate FH model structures. Permatasari and Ubaidillah (2021) document their implementation in the R package `msae`.

The second is the sampling-error covariance matrix of the direct estimators. This matrix is treated as known by the MFH model and supplied as part of the input data. It describes uncertainty from the survey design, not unexplained area heterogeneity.

This note concerns the second object: the sampling-error covariance matrix of the direct survey estimators.

3. Object to be estimated

For area j and period t, define the Fay–Herriot sampling error as

$$e_{jt} = \hat{Y}_{jt} - \theta_{jt},$$

where $\hat{Y}_{jt}$ is the direct survey estimate and θ_{jt} is the true area value. For two adjacent periods, the off-diagonal target is

$$Cov\left(e_{jt}, e_{j,t+1}\right).$$

From a design-based survey-sampling perspective, θ_{jt} is fixed. Therefore,

$$Cov\left(e_{jt}, e_{j,t+1}\right) = Cov\left(\hat{Y}_{jt}, \hat{Y}_{j,t+1}\right).$$

The MFH sampling-error covariance is therefore the design-based covariance of the direct survey estimators. Särndal, Swensson, and Wretman (1992) provide the general survey-sampling basis for weighted direct estimators and their variances and covariances; Binder (1983) provides the basis for Taylor-linearized variance estimation for estimators from complex surveys.

4. Household-level intuition

The attached internal notes motivate the covariance using the household-level model

$$y_{hjkt} = X_{hjkt}\beta + u_{jk} + \varepsilon_{hjkt},$$

where h indexes households, k indexes PSUs, j indexes areas, and t indexes time. In this representation:

- u_{jk} is a persistent PSU or cluster effect;
- household-level idiosyncratic errors are not serially correlated;
- covariates are assumed not to create cross-time covariance;
- the persistent cluster effect is what generates sampling-error covariance across years.

The implication is practical. If the same PSU appears in two survey waves, households in that PSU share a persistent cluster component. Therefore, even different households in the same repeated PSU can contribute to cross-time covariance. This project-specific derivation is from Yoshida (2025a, 2025b), not from Benavent and Morales (2016).

5. Three equivalent covariance expressions

The following three expressions summarize the derivation in Yoshida (2025a). For readability, area j is suppressed in the notation, and the two waves are labeled 1 and 2.

A. Household form

Let S_{12} be the set of m PSUs appearing in both waves. Let H_{k1} and H_{k2} be the sampled households in PSU k in waves 1 and 2. Let w_{hk1} and w_{lk2} be the final survey weights, and let N_1 and N_2 be the population totals implied by the weights. Then

$$Cov\left(\hat{\bar{Y}}_1, \hat{\bar{Y}}_2\right) = \frac{1}{N_1 N_2} \sum_{k \in S_{12}} \sum_{h \in H_{k1}} \sum_{l \in H_{k2}} w_{hk1} w_{lk2} Cov\left(Y_{hk1}, Y_{lk2}\right).$$

This expression makes clear that the covariance is not limited to households observed in both waves. Under a persistent PSU effect, cross-household pairs inside a repeated PSU also contribute.

B. PSU-population form

Define weighted PSU totals:

$$T_{k1} = \sum_{h \in H_{k1}} w_{hk1} Y_{hk1}, T_{k2} = \sum_{l \in H_{k2}} w_{lk2} Y_{lk2}.$$

Grouping the household terms within PSU k gives

$$Cov\left(T_{k1}, T_{k2}\right) = \sum_{h \in H_{k1}} \sum_{l \in H_{k2}} w_{hk1} w_{lk2} Cov\left(Y_{hk1}, Y_{lk2}\right).$$

Thus the covariance of the two direct means can be written as a covariance of PSU totals:

$$Cov\left(\hat{\bar{Y}}_1, \hat{\bar{Y}}_2\right) = \frac{m}{N_1 N_2} Cov_k\left(T_{k1}, T_{k2}\right).$$

C. PSU-sample form

Replacing the PSU-population covariance with its unbiased sample estimator yields

$$\widehat{Cov}\left(\hat{\bar{Y}}_1, \hat{\bar{Y}}_2\right) = \frac{m}{m-1}\frac{1}{N_1 N_2}\sum_{k \in S_{12}} \left(T_{k1} - \acute{T}_1\right)\left(T_{k2} - \acute{T}_2\right),$$

where

$$\acute{T}_t = \frac{1}{m}\sum_{k \in S_{12}} T_{kt}.$$

The household form becomes the PSU form by grouping all household-pair covariance terms within each PSU and using the bilinearity of covariance. The PSU-population form becomes the PSU-sample form by replacing the PSU covariance with its finite-sample estimator. Yoshida (2025a) provides these three equivalent expressions. Särndal, Swensson, and Wretman (1992) and Binder (1983) are the general survey-sampling references for the underlying design-based variance and covariance logic.

6. Why R's `survey` package estimates the right quantity

When the survey design is specified with ids = ~psu, Taylor linearization treats each PSU total as a single first-stage observation. The covariance returned by

```
svyby(..., covmat = TRUE)
vcov(...)
```

therefore corresponds to the PSU-sample covariance expression above, after scaling totals to means. In operational terms:

> With PSUs declared as first-stage units, the ordinary design-based covariance returned by R's `survey` package estimates the Fay–Herriot sampling-error covariance needed by the MFH model. No special household-level covariance estimator is required.

Binder (1983) supports the use of Taylor-linearized variance estimation for estimators from complex surveys. Lumley (2004, 2010) documents the R `survey` package and its implementation of complex-survey analysis. Yoshida (2025a, 2025b) provides the specific link between the `survey` package output and the MFH sampling-error covariance in this package context.

The same principle applies in Stata or other survey software when the survey design is correctly specified and PSUs are treated as first-stage sampling units.

In principle, then, the off-diagonal of the area matrix is the per-area design-based covariance returned by `svyby(..., covmat = TRUE)`. In practice, area-level covariances estimated from only a few overlapping clusters are highly unstable, and the resulting two-by-two matrices are often erratic or not positive definite. By default the package therefore replaces the per-area off-diagonal with a national-correlation reconstruction (Section 8), while still offering the `direct` per-area covariance as an option.

7. Warning against the panel-only shortcut

Estimating covariance using only re-interviewed households is generally wrong.

The panel-only shortcut implicitly assumes

$$Cov\left(Y_{hk1}, Y_{lk2}\right) = 0 \, for \;\; all \;\; h \neq l.$$

But under the persistent PSU-effect model, households in the same PSU share the same cluster effect. For households in the same repeated PSU,

$$Cov\left(Y_{hk1}, Y_{lk2}\right) = \sigma_u^2$$

whether $h = l$ or $h \neq l$. Therefore, all cross-household pairs within a repeated PSU contribute to covariance, not only households observed in both waves. Dropping non-panel households discards most of the covariance contribution whenever the persistent PSU effect is material.

The covariance should be estimated using the full sample across both waves, with PSUs correctly declared, not only the linked panel households.

This warning follows Yoshida (2025a, 2025b). It should not be attributed to Benavent and Morales (2016), whose role here is the MFH model requiring a sampling-error covariance input.

8. Practical implementation in the EU SAE package

The two-year covariance implementation in scripts/02_mfh.R uses survey::svydesign with ids = ~psu, optional supplied strata, and weights = ~population_weight. For household rows, population_weight = weight × household size; for person rows it equals weight alone. Domain covariance comes from the correctly specified joint survey design via svyby(..., covmat = TRUE) and vcov(). A national cross-time correlation can be combined with the chosen domain variances.

Estimating a separate sampling-error covariance for every area is rarely reliable. Many areas contain only a handful of clusters that overlap across the two waves, so this per-area design covariance is noisy and the resulting two-by-two matrix can even fail to be positive definite. To keep the model input stable, the package simplifies the off-diagonal: it treats the cross-time correlation of the sampling errors as common across areas, estimates that single correlation once at the national level, following Boonstra and van den Brakel (2019), and rebuilds each off-diagonal from the sampling variances of the same area. For area j the sampling-error matrix then takes the form

$$D_j = \begin{pmatrix} \hat{V}_{j1} & \hat{\rho}\sqrt{\hat{V}_{j1}\hat{V}_{j2}} \\ \hat{\rho}\sqrt{\hat{V}_{j1}\hat{V}_{j2}} & \hat{V}_{j2} \end{pmatrix}.$$

Estimating the national cross-time autocorrelation

The shared correlation is the cross-time autocorrelation of the national direct estimator. The package fits the national year means on the same complex design (`ids = ~psu`), takes the two-by-two national

design covariance returned by `svyby(..., covmat = TRUE)`, and converts it to a correlation matrix with `cov2cor()`. The off-diagonal of that correlation matrix is the estimate

$$\hat{\rho} = \frac{\widehat{Cov}\left(\hat{Y}_{\cdot 1}, \hat{Y}_{\cdot 2}\right)}{\sqrt{\widehat{Var}\left(\hat{Y}_{\cdot 1}\right)\widehat{Var}\left(\hat{Y}_{\cdot 2}\right)}}.$$

Computed from the full national sample, this correlation is far more stable than any single area covariance. Applying it to the area-level variances gives the off-diagonal

$$\widehat{Cov}\left(e_{j1}, e_{j2}\right) = \hat{\rho}\sqrt{\hat{V}_{j1}}\sqrt{\hat{V}_{j2}}.$$

where the two variances under the square roots are the diagonal sampling variances chosen for area j, described next.

Variance choices on the diagonal

The same variance concept must scale the diagonal and off-diagonal. direct keeps raw design variances subject to explicit missing/zero safety backfills. sm_all is the default and replaces all working variances with GVF predictions. Optional sm_out retains a finite direct variance at least 0.001 and replaces missing, nonfinite or strictly smaller values with GVF predictions; it does not replace large variances. This cutoff is an application policy requiring indicator/scale sensitivity checks. For log means it applies on the log-mean variance scale. Raw Direct uncertainty remains separate from these model inputs.

Smoothing uses a generalized variance function: the log of the direct variance is regressed on the log of the area sample size, and the fitted value is back-transformed with a lognormal correction,

$$\log \hat{V}_{jt}^{dir} = \alpha + \beta \log n_{jt} + \varepsilon_{jt}, \hat{V}_{jt}^{sm} = \exp\left(\hat{\alpha} + \hat{\beta}\log n_{jt}\right)\exp\left(\hat{\sigma}^2/2\right).$$

These variance choices line up with the covariance options below, which set how the off-diagonal itself is formed.

Within this structure, the package exposes five options for the off-diagonal of the sampling-error matrix:

- `direct`: use direct design-based domain-level covariance from the survey variance-covariance matrix.
- `rho_dir`: estimate a national cross-time correlation and apply it to raw direct domain variances.
- `rho_sm_out`: estimate a national cross-time correlation and apply it to selectively smoothed (sm_out) domain variances.
- `rho_sm_all`: estimate a national cross-time correlation and apply it to fully smoothed domain variances.
- `zero`: assume zero cross-time covariance.

Recommended use:

- Use `direct` when domain-level covariance estimates are stable and there are enough PSUs per area.

- Use rho_sm_all with sm_all as the default when sampling dependence is plausible and domain covariances are noisy; assess the GVF fit and covariance assumptions.
- Use rho_sm_out with sm_out as an optional sensitivity. Its unchanged 0.001 cutoff is scale-specific and does not replace large variances.
- Use `rho_dir` when raw direct variances are trusted.
- Use `zero` only when samples are genuinely independent across years, or as a documented sensitivity scenario.

These options are EU SAE package implementation choices informed by the attached internal notes. They are not direct prescriptions from Benavent and Morales (2016), Lumley (2004), or Särndal, Swensson, and Wretman (1992).

The covariance option should be consistent with the variance option used on the diagonal of the sampling-error covariance matrix. For example, `rho_sm_out` uses the national cross-time correlation with the selectively smoothed (sm_out) diagonal variances, while `rho_sm_all` uses the fully smoothed diagonal variances. This consistency helps avoid covariance matrices where the off-diagonal was built from one variance concept while the diagonal uses another.

The dashboard offers matching variance–covariance choices (sm_all with rho_sm_all, sm_out with rho_sm_out, and direct with rho_dir), with direct and zero covariance also available. New configurations default to sm_all with rho_sm_all. An omitted covariance setting is derived from the effective variance choice; explicit saved choices are preserved. UFH also honors the selected variance option with log mean welfare, on the log-mean variance scale. Arcsine UFH retains its separate effective-sample-size analytical variance, with a fallback to observed sample size when its estimated effective sample size is invalid.

For log mean welfare, individual zero and negative welfare contributes to the arithmetic Direct mean; positive finite area means are required before logging. The delta transformation applies to the full covariance matrix. Any admissibility repair is performed once before the original MFH fit, recorded in mfh_sampling_covariance.csv and held fixed throughout the bootstrap. GVF and estimated correlation uncertainty are conditional inputs, not additional resampled components.

Relation to Boonstra and van den Brakel (2019)

Building the cross-time covariance from a single nationally-pooled correlation is the strategy of Boonstra and van den Brakel (2019), who estimate the sampling-error covariances of the Dutch Labour Force Survey as inputs to a structural time series model. For two estimates of the same area their covariance takes the form

$$\widehat{Cov}\left(\hat{Y}_{i1}, \hat{Y}_{i2}\right) = \frac{n_{i,12}}{n_{i1} n_{i2}} \hat{\rho}^{unit} \sqrt{v\left(\hat{Y}_{i1}\right) v\left(\hat{Y}_{i2}\right)}.$$

where the leading fraction is the share of sample units observed in both periods, and the correlation is taken between the unit-level regression residuals on those matched units, pooled over areas. Because the Labour Force Survey is a rotating panel that re-interviews the same persons, the cross-time dependence is carried by these matched units. Boonstra and van den Brakel note that within-household clustering can be ignored in their setting because the intra-cluster correlation of unemployment is small.

The package retains a nationally pooled correlation, estimated with the survey design and PSUs declared as first-stage units, and applies it to area-specific working variances without an additional household-overlap factor. This is a project implementation choice. Its adequacy depends on consistent PSU identifiers, survey design and the plausibility of a common correlation; it should be checked against design information and sensitivity results.

- Welfare may be geographically clustered, making persistent PSU-level dependence plausible. Its magnitude and sign must be checked for the application using the joint design and population evolution. Shared PSUs can matter beyond re-interviewed households; a matched-unit correlation alone need not capture all relevant dependence. This is an application-specific expectation, not a universal large-correlation claim.
- Under a persistent cluster-effect model, reused PSUs can induce covariance even without re-interviewing the same households. Its magnitude and sign depend on the actual design and population; PSU reuse alone is not an empirical covariance estimate.
- It needs only consistent PSU identifiers, not household microdata linked across rounds. Survey systems used for poverty mapping routinely retain PSUs from a master sample but rarely allow households to be matched across years.
- It is coherent with the diagonal. The variances are already estimated with PSUs as first-stage units, so estimating the off-diagonal at the same level keeps the whole matrix on one design basis and uses the same `survey` package machinery.

This is an adaptation rather than a correction. The unit-level correlation of Boonstra and van den Brakel (2019) is the natural quantity for a rotating panel in which the dependence comes from re-interviewing the same people and within-household clustering is weak, and their model also resolves a far finer wave- and month-specific autocorrelation structure than the single cross-time correlation used here. For cross-sectional poverty mapping built on PSU-reuse designs, estimating the same nationally-pooled correlation at the PSU level is the better-suited and more widely applicable form.

9. Convergence and stability

Including non-zero covariance can create convergence difficulties in MFH estimation, especially when domain-level covariance estimates are noisy, outlying, or lead to near-singular sampling-error covariance matrices. This is a numerical and diagnostic issue, not an automatic reason to ignore covariance.

Practical remedies include:

- using a national-correlation approach rather than noisy domain-specific covariances;
- smoothing variances and covariances consistently;
- replacing only clear covariance outliers with smoothed values;
- applying small discounting to covariances only as a numerical sensitivity check, not as a substantive assumption;
- using zero covariance only when justified by the survey design or as a sensitivity scenario;
- documenting any adjustment to the covariance matrix.

For a two-year matrix, analysts should check that

$$\left|Cov\left(e_{j1}, e_{j2}\right)\right| < \sqrt{Var\left(e_{j1}\right) Var\left(e_{j2}\right)}$$

apart from small numerical tolerances. Otherwise, the supplied sampling-error matrix is not positive definite. When a covariance estimate violates this condition, the analyst should treat it as a diagnostic problem, not silently force the model to run.

This section is practical implementation guidance from the EU SAE package and Yoshida (2025a, 2025b). It is not a direct conclusion of the published MFH literature.

10. Implications for change over time

The covariance of direct estimators affects the variance of the estimated change. For two years,

$$Var\left(\hat{Y}_{j,t+1} - \hat{Y}_{jt}\right) = Var\left(\hat{Y}_{j,t+1}\right) + Var\left(\hat{Y}_{jt}\right) - 2\,Cov\left(\hat{Y}_{j,t+1}, \hat{Y}_{jt}\right).$$

Ignoring relevant covariance can produce misleading confidence intervals for changes. If the covariance is positive, ignoring it usually overstates the direct estimator variance of a difference. If a package or analyst later propagates uncertainty through an MFH model, the effect also depends on the fitted MFH model, the supplied cap D sub j matrices, and the prediction-error covariance of the final estimates.

For MFH-based changes, valid inference requires attention to two layers:

- the sampling-error covariance matrix supplied to the MFH model;
- the prediction-error covariance or MCPE of the final model-based estimates.

Benavent and Morales (2016) present the concepts of multivariate FH MSE and MCPE. Yoshida (2025a, 2025b) motivates the role of the cross-time sampling-error covariance in the EU SAE implementation.

11. Practical checklist

Before estimating an MFH model, analysts should confirm that:

- PSU identifiers are available and consistently coded across years;
- the survey design object includes the full sample from both waves;
- panel-only households are not used as the sole basis for covariance estimation;
- direct estimates and variances have been computed by area and year;
- the cross-time covariance option matches the survey design;
- diagonal variances and off-diagonal covariances are based on consistent variance choices;
- resulting covariance matrices are positive definite or adjusted transparently;
- sensitivity checks have been run using plausible alternative covariance assumptions;
- the selected covariance option is documented in the final report.

12 Implementation workflow

The following workflow summarizes the current production code; it is not a standalone executable listing.

The illustrative R workflow is: construct population_weight = weight × household size for household rows, or population_weight = weight for person rows; declare the supplied PSU and optional strata design with survey::svydesign; compute arithmetic Direct means or FGT indicators by domain and year with survey::svyby; obtain cross-year design covariance from the joint design; and apply the selected area-variance and correlation policy. The production implementation is scripts/02_mfh.R. For log means, transform only after arithmetic aggregation: variance V/mean^2 and covariance $C/(\text{mean1} \times \text{mean2})$. Validate the assembled covariance matrices once before fitting, retain the repaired matrices, and use those exact inputs for every bootstrap simulation and refit.

Lumley (2004, 2010) is the software reference for the `survey` package. Yoshida (2025a, 2025b) provides the project-specific explanation of why this covariance corresponds to the MFH sampling-error covariance.

13. Reference list

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14. Citation boundaries

This note uses the sources as follows:

- Fay and Herriot (1979): original area-level Fay–Herriot model.
- Rao and Molina (2015): general small area estimation framework, direct estimators, EBLUPs, sampling variances, MSE, and applied interpretation.
- Benavent and Morales (2016): multivariate FH model, sampling-error covariance matrix as a model input, REML/EBLUP fitting, MSE, and MCPE concepts.
- Permatasari and Ubaidillah (2021): `msae` software implementation of MFH model variants.
- Särndal, Swensson, and Wretman (1992): design-based survey sampling theory, weighted estimators, and variance/covariance estimation.
- Binder (1983): Taylor-linearized variance estimation for complex surveys.
- Lumley (2004, 2010): R `survey` package and complex-survey implementation.
- Boonstra and van den Brakel (2019): origin of the nationally-pooled autocorrelation strategy for estimating cross-time sampling-error covariance, using a unit-level (rotating-panel) correlation.
- Yoshida (2025a, 2025b): project-specific derivation linking design-based survey covariance to the FH sampling-error covariance, the PSU-level adaptation of the Boonstra and van den Brakel (2019) strategy, and the warning that panel-only covariance estimation can understate covariance.